

# Rangaswamy A

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## SUMMARY

Aspiring AI/ML Engineer with hands-on experience building practical AI solutions. Passionate about applying machine learning and artificial intelligence to solve real-world problems and develop practical applications.

## SKILLS

- **Programming Languages:** Python, SQL
- **Machine Learning:** Supervised & Unsupervised Learning, Ensemble Learning, Model Evaluation, Regularization (Lasso, Ridge), Overfitting & Underfitting
- **Machine Learning Algorithms:** Linear Regression, Logistic Regression, Decision Trees, SVM, KNN, Naive Bayes, K-Means, DBSCAN, PCA, Hierarchical, Isolation Forest, LOF
- **Deep Learning:** ANN, FNN, CNN, RNN, LSTM, Transformers
- **Generative AI:** LLMs, NLP, RAG, AI Agents, OpenAI API, GAN
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Web Technologies:** HTML, CSS
- **Frameworks & Tools:** Scikit-learn, PyTorch, FastAPI, Streamlit
- **Developer Tools:** Git, GitHub, Docker, CI/CD

## EDUCATION

- B.E. in Artificial Intelligence & Machine Learning | ACET College | CGPA: 7.63 |  
Currently Pursuing 7th Semester
- XII (DPUE) | Govt. Ex-Municipal PU College 86.5% | 2023

## ACADEMIC PROJECTS

### AI-Powered Cyber Threat Detection System — Individual Project

- Developed a network detection system using Logistic Regression and XGBoost, preprocessing 8 network traffic datasets and selecting the top 60 features for classification.
- Evaluated the model on completely unseen Friday network traffic data to validate its ability to generalize to new network traffic patterns.
- Built and deployed a Streamlit web application for CSV-based threat detection with attack probability, batch predictions, detection summaries, and downloadable results.
- Technologies: Python, Pandas, NumPy, Scikit-learn, XGBoost, Streamlit, Joblib

[GitHub](#) | [Live Demo](#)

### AI-Powered Document Intelligence RAG — Individual Project

- Developed an end-to-end RAG-based document Q&A system that processes uploaded PDFs and generates context-grounded answers using an LLM.
- Implemented PDF extraction, text chunking, Hugging Face embeddings, FAISS semantic search, and relevance checking with source-page retrieval.
- Built and deployed an interactive Streamlit application supporting PDF upload, document statistics, conversational Q&A, chat history, and multiple PDF documents.
- Technologies: Python, Streamlit, LangChain, Groq LLM, FAISS, Hugging Face Embeddings, PyPDF, RAG, Semantic Search.

[GitHub](#) | [Live Demo](#)

### AI YouTube Video Analyzer — Individual Project

- Developed an AI-powered YouTube Video Analyzer using Agno AI Agent and Groq LLM to generate video summaries, detailed analysis, study notes, key takeaways, quizzes, and timestamp-based insights from YouTube videos.
- Built an interactive Streamlit application with YouTube metadata extraction, AI-powered content analysis, and downloadable PDF/Text reports for improved learning and content understanding.
- Technologies: Python, Streamlit, Agno AI, Groq LLM, yt-dlp, YouTube Tools, ReportLab.

[GitHub](#) | [Live Demo](#)

**Live Demo Note:** Applications are deployed on Streamlit Community Cloud. If an app is inactive, click “Yes, get this app back up!” to wake it up; the application will then load normally.

## CERTIFICATION

Prime (AI/ML) — Certificate of Completion, Apna College | [\[View Certificate\]](#)